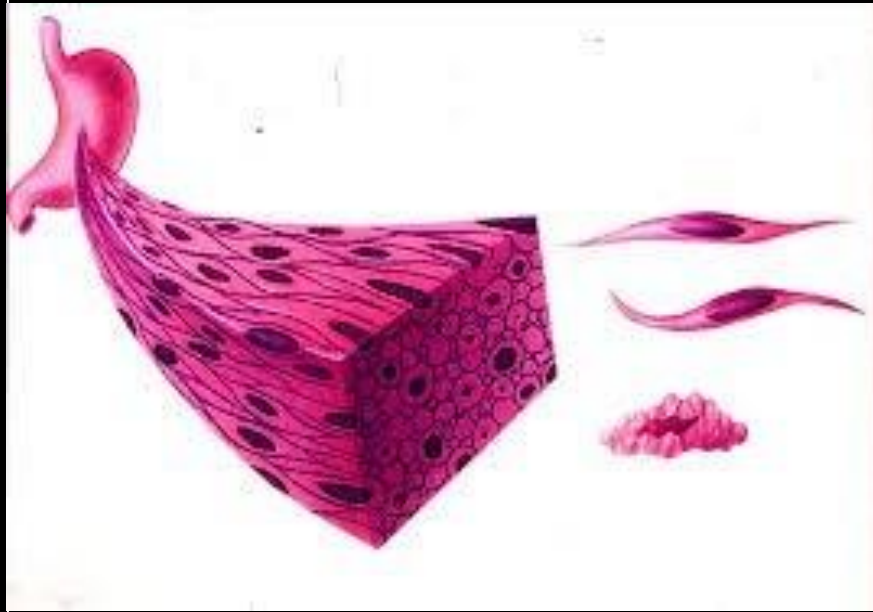




SMOOTH MUSCLE PHYSIOLOGY.

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MD, AFMSA,
DMHS
ASSOCIATE PROF
DEPT. OF PHYSIOLOGY**

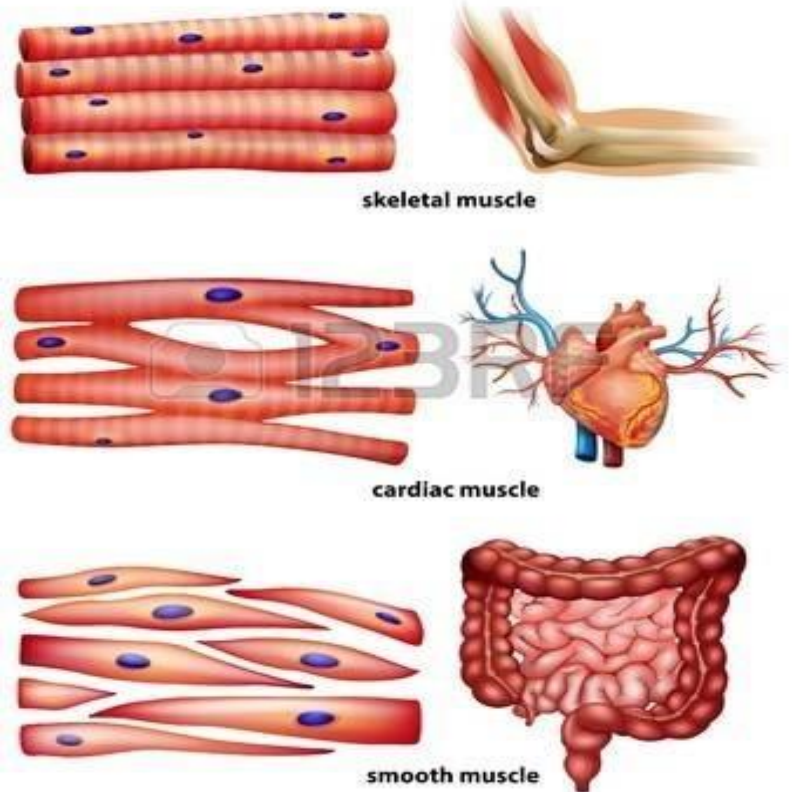
OBJECTIVES.

- ✓ Functional Anatomy & Organization
- ✓ Process of Excitability & Contraction.
- ✓ Characteristics of Excitability & Contractility.
- ✓ Excitation & Inhibition.

FUNCTIONAL ANATOMY & ORGANIZATION

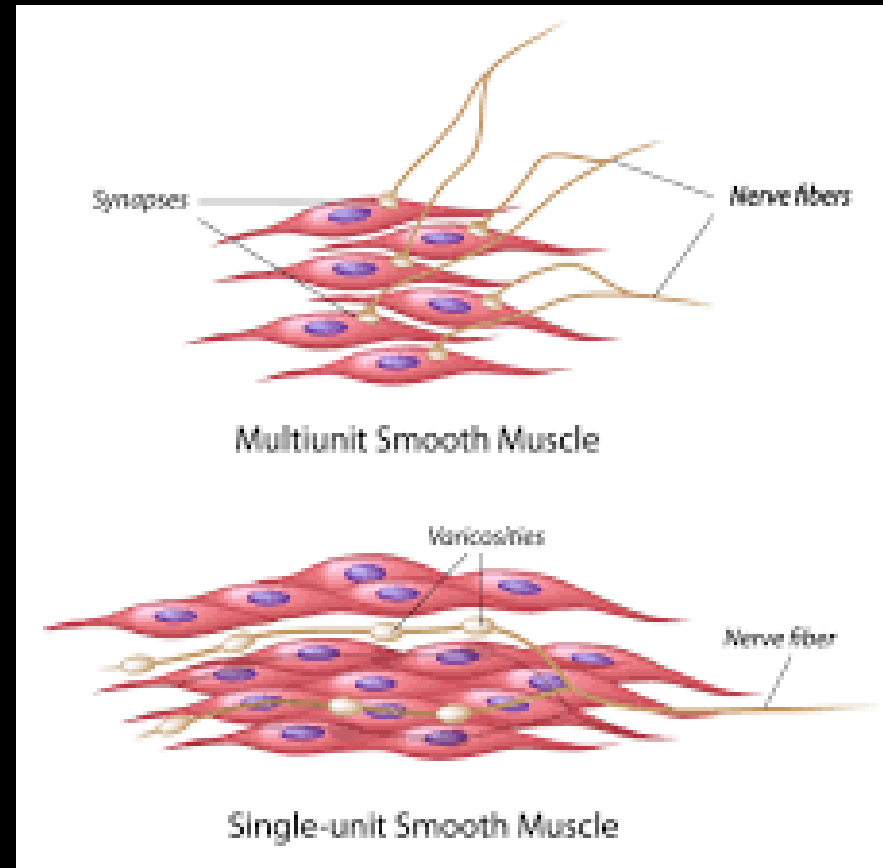
- ✓ Non-striated Muscle.
- ✓ Involuntary muscle.
- ✓ Long fusiform in bundles or fasciculi.

Types of Muscle Cells



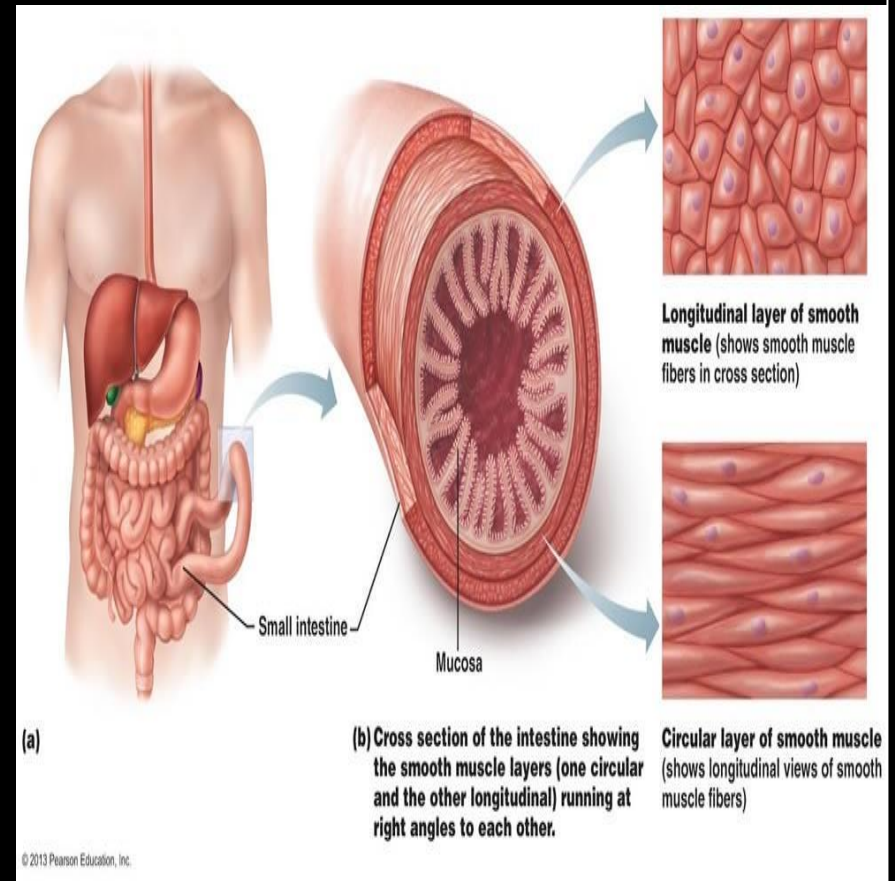
TYPES OF SMOOTH MUSCLES

- ✓ Single unit smooth muscle
- ✓ Multi unit smooth muscle.



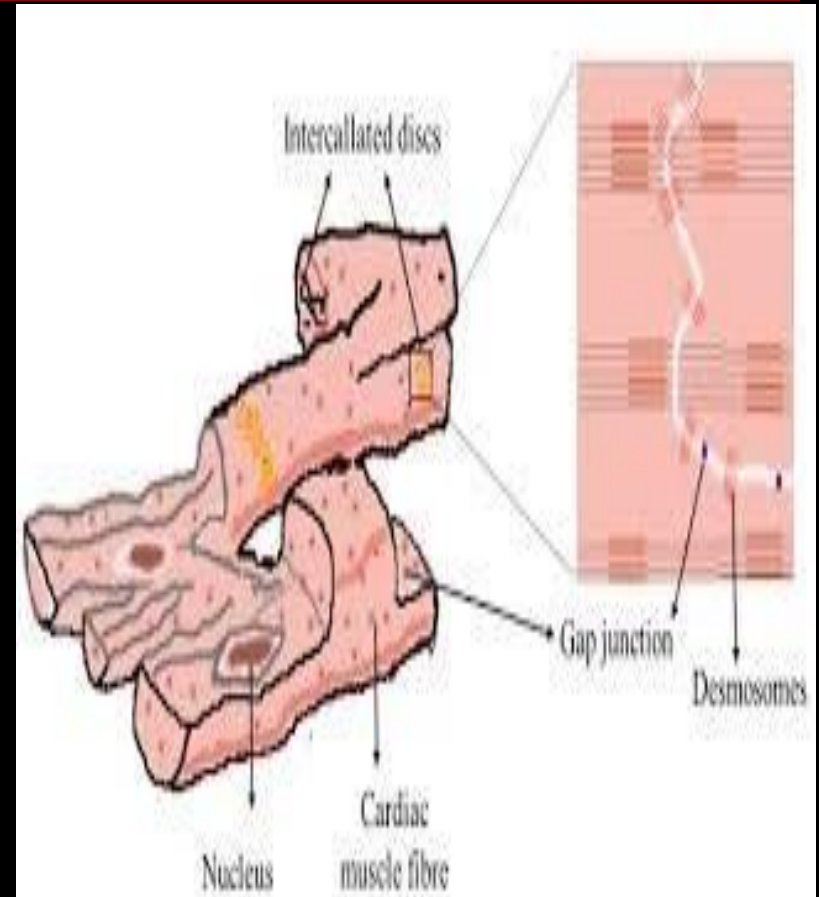
SINGLE UNIT SMOOTH MUSCLE

- ✓ **Visceral** smooth muscles
- ✓ Present in hollow viscera like GIT, Uterus, Ureter, Urinary Bladder & Respiratory tract.



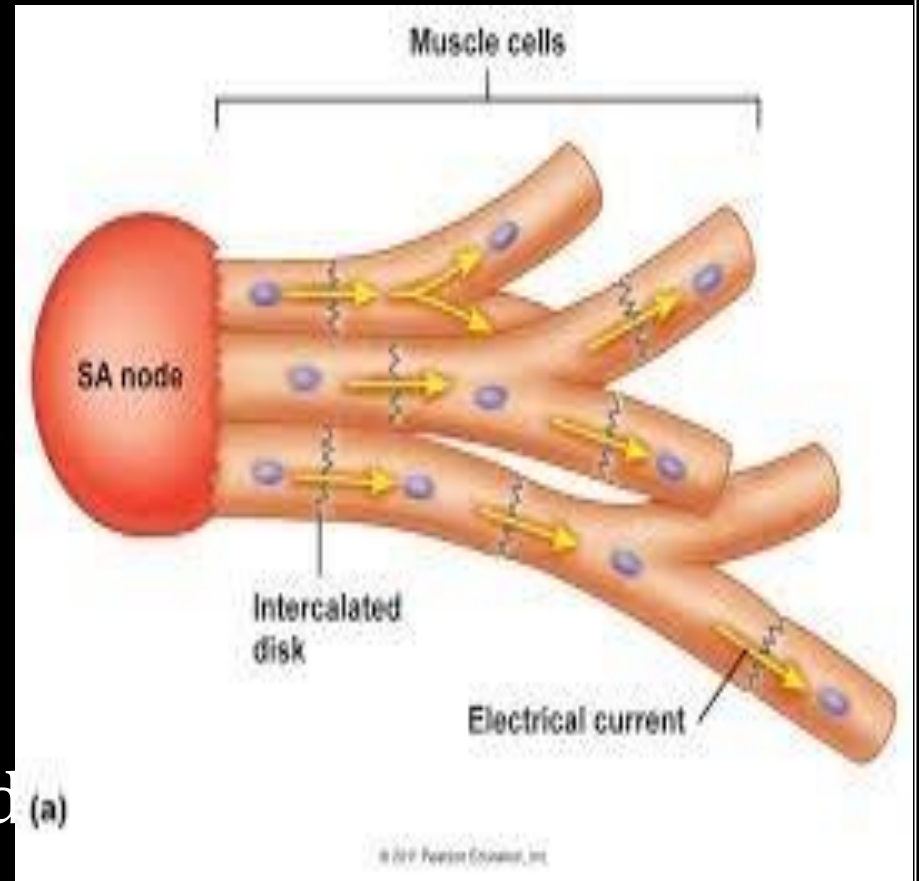
SALIENT FEATURES

- ✓ Arranged in large sheet.
- ✓ Has low resistance bridges (**Gap Junctions**) between individual muscles.
- ✓ Function in a **Syncytial** fashion & contract as a single unit.
- ✓ Has their own Rhythmic contractility (**Myogenic Tone**)



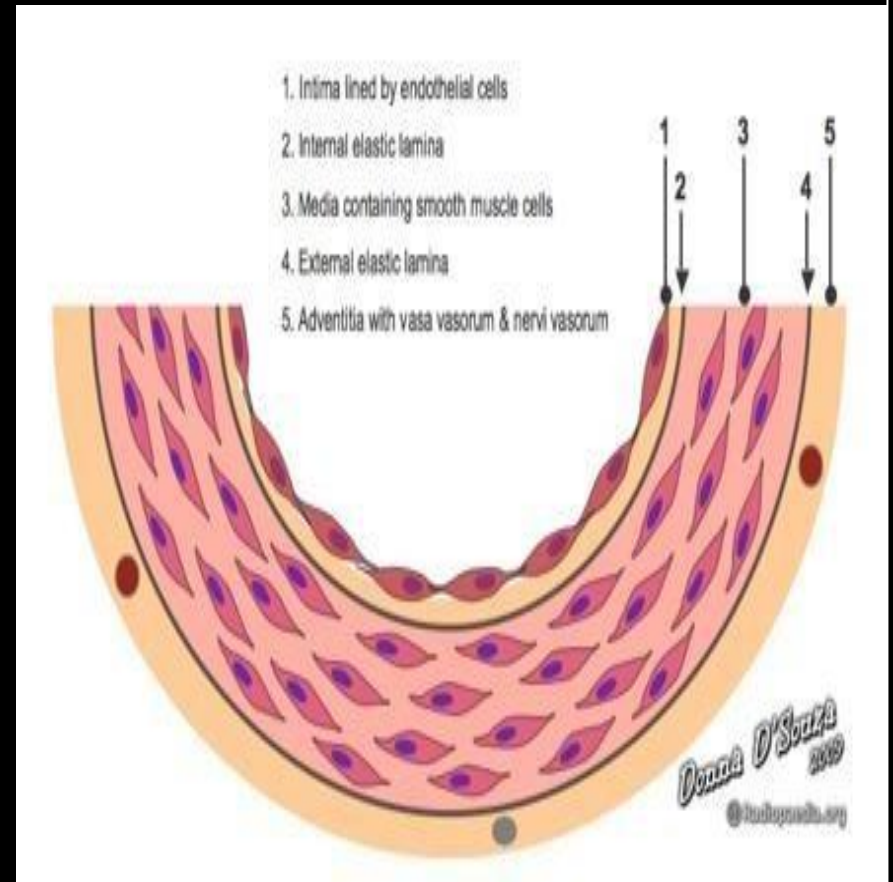
SALIENT FEATURES

- ✓ Rate of contraction determined by **Pacemaker** region & **Nerve** supply only **Modulates** its activity.
- ✓ Contraction is stimulated by **Stretching** involved in **Autoregulation** of blood flow.



MULTI UNIT SMOOTH MUSCLES

- ✓ Made up of Multiple units without interconnecting bridges-**Non-Syncytial**.
- ✓ **Location** – Blood vessels, Epididymis, Vas deference, Iris, Ciliary body & Piloerector muscle.



SALIENT FEATURES

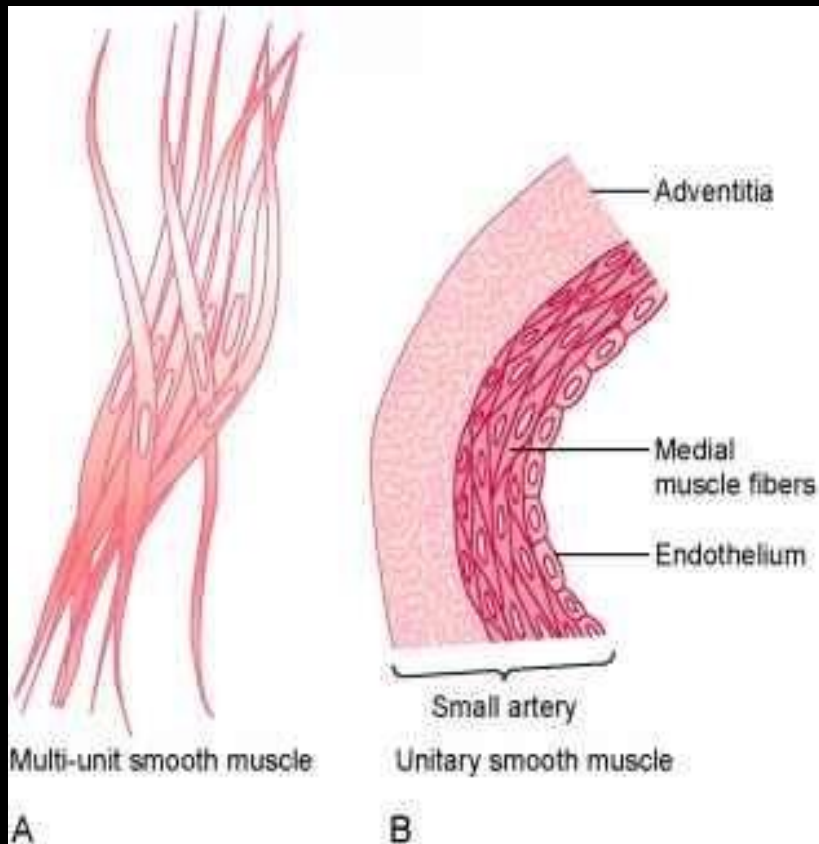
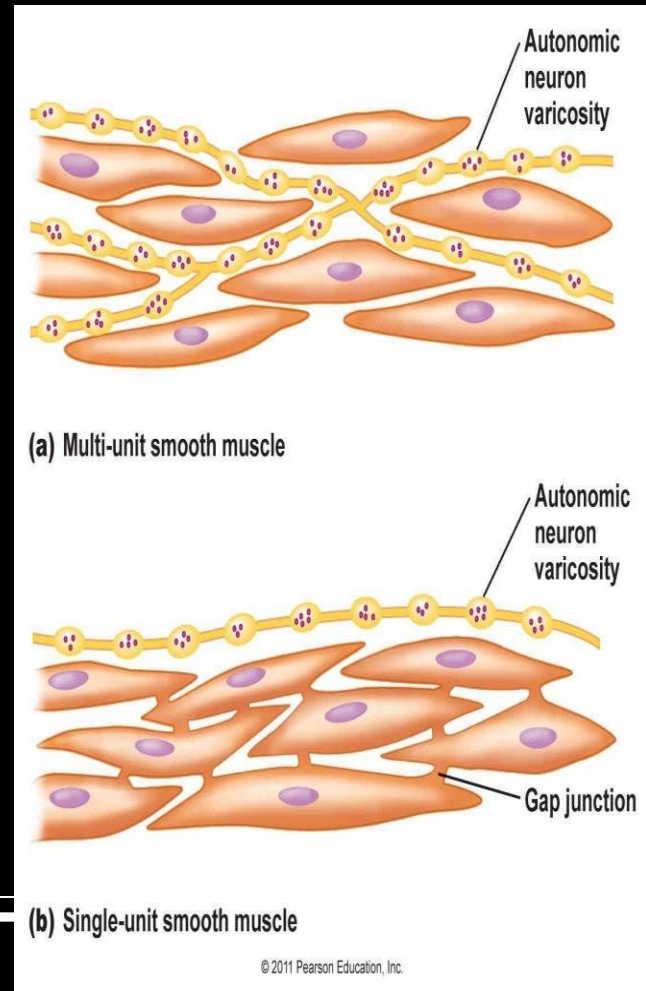


Figure 8-1

- ✓ Multi-unit muscle each innervated by single nerve ending.
- ✓ Contraction is **Neurogenic** & each stimuli causes irregular **Tetanic** contraction.

SALIENT FEATURES

- ✓ Do not show spontaneous contraction – **No Pacemaker activity.**
- ✓ **No Gap Junction** so excitation localized within motor unit.
- ✓ **Do not** respond to stretch.



Innervations of smooth muscles

- ✓ **Nerve supply** –
Autonomic both
Sympathetic &
Parasympathetic.
- ✓ **Symp** – Contraction &
Parasymp – Relaxation.
- ✓ Preganglionic fibres –
ganglion – postganglionic
fibres – muscle.

Autonomic Output Pathways

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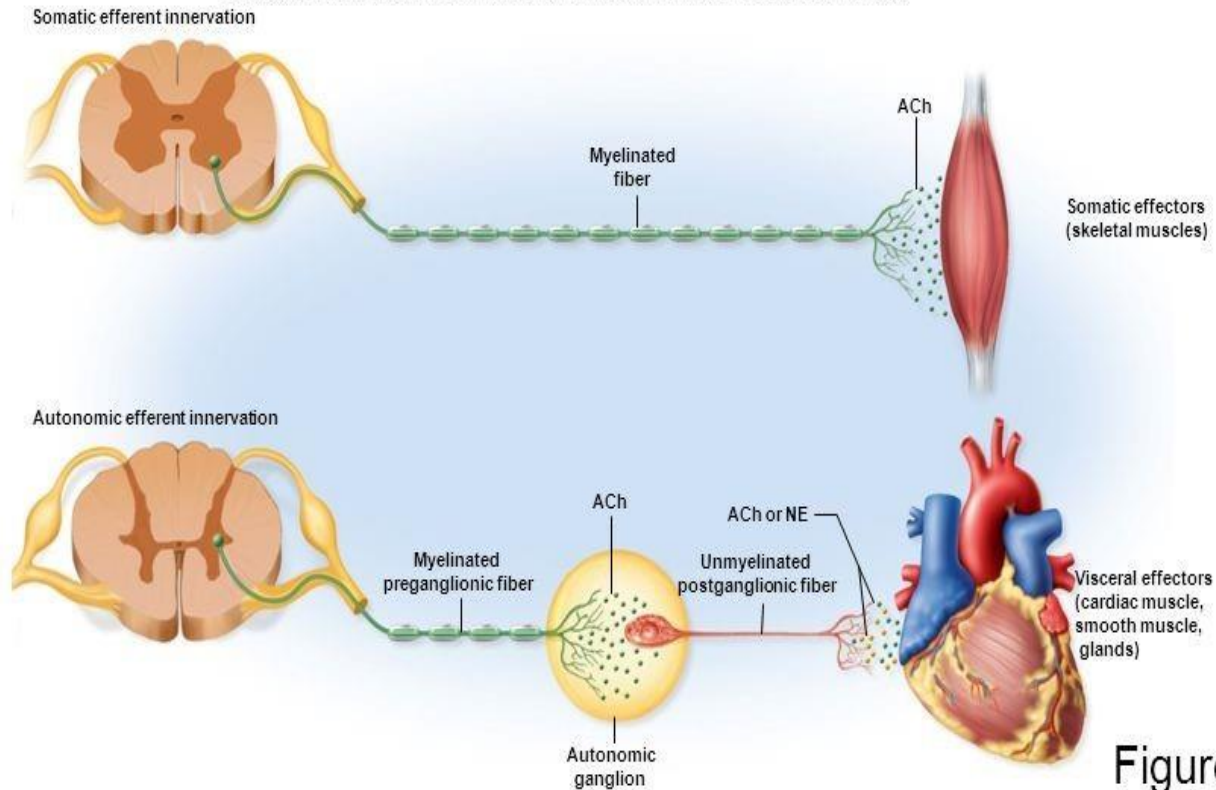


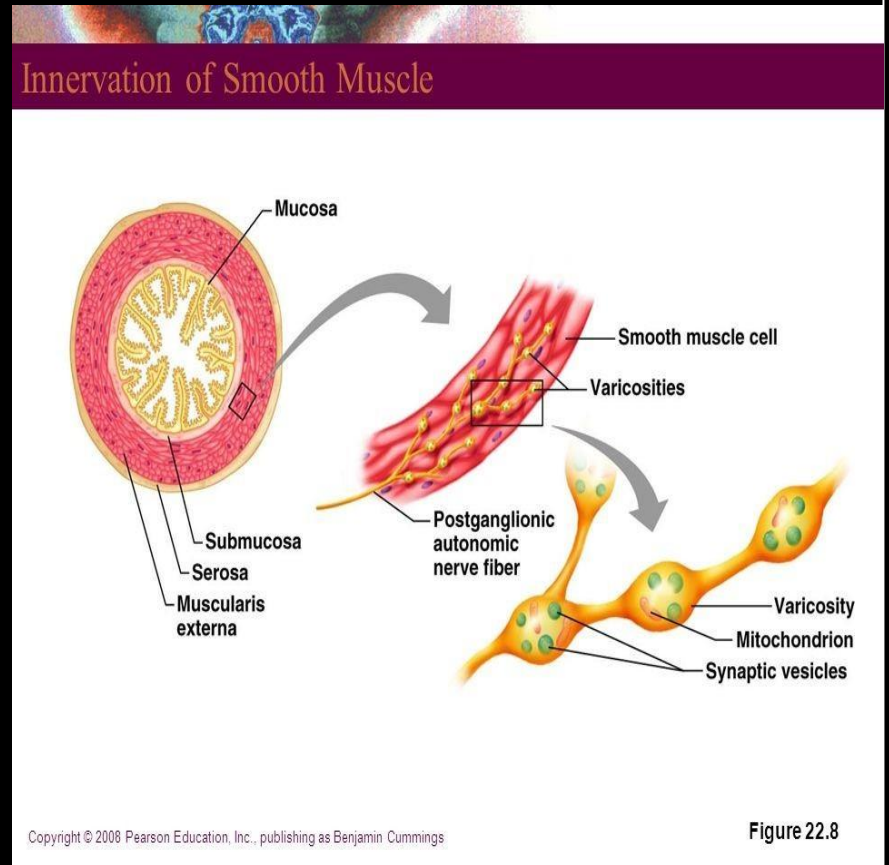
Figure 15.2

ANS—two neurons from CNS to effectors

- Presynaptic neuron cell body is in CNS
- Postsynaptic neuron cell body is in peripheral ganglion

Neuromuscular junction of Smooth Muscles

- ✓ Post ganglionic fibres branch extensively.
- ✓ Neuronal network has **Beaded** appearance due to varicosities.
- ✓ **Varicosities** contains chemical neurotransmitter (**Ach/NE**)

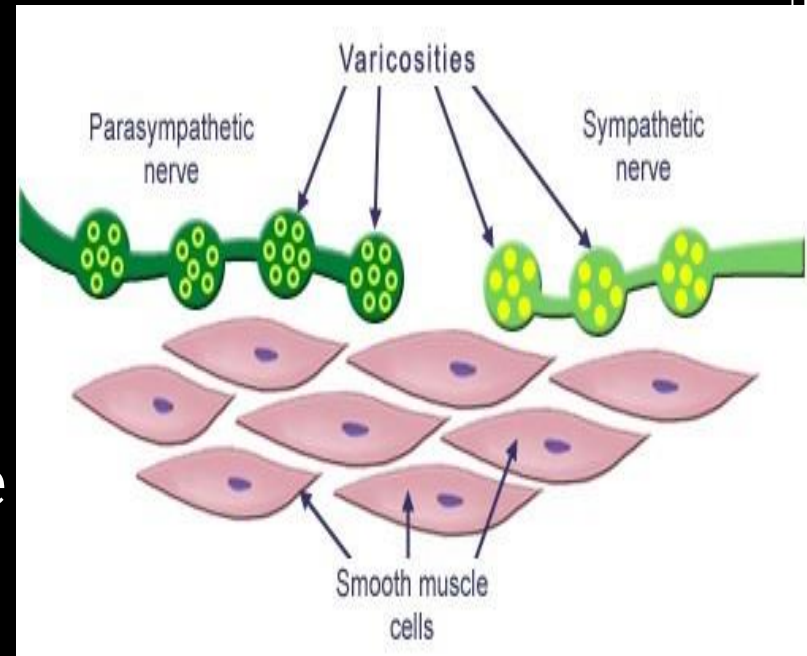


Neuromuscular junction of smooth muscles

Nerve fibres **do not end in Motor End Plate** but releases its neuro-transmitter in **interstitial** fluid near muscle fibre.

It then **diffuses** in muscle fibre & causes activation.

So single stimuli will not cause activation of entire muscle
Multiple Stimuli are required.

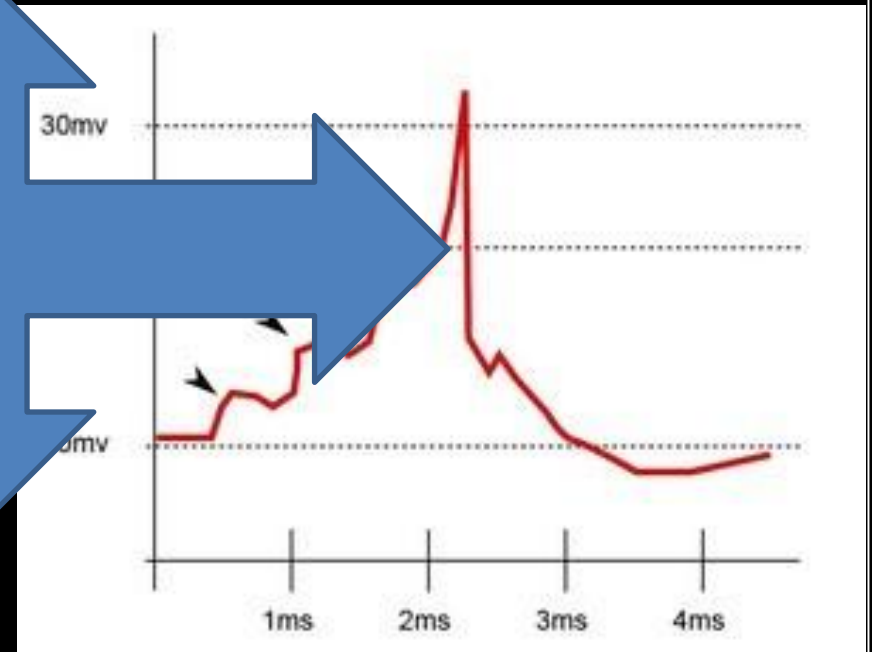


Excitatory Junctional Potential (EJP) or Inhibitory Junctional potential (IJP)

- Either depolarizing or repolarizing responses are recorded in response

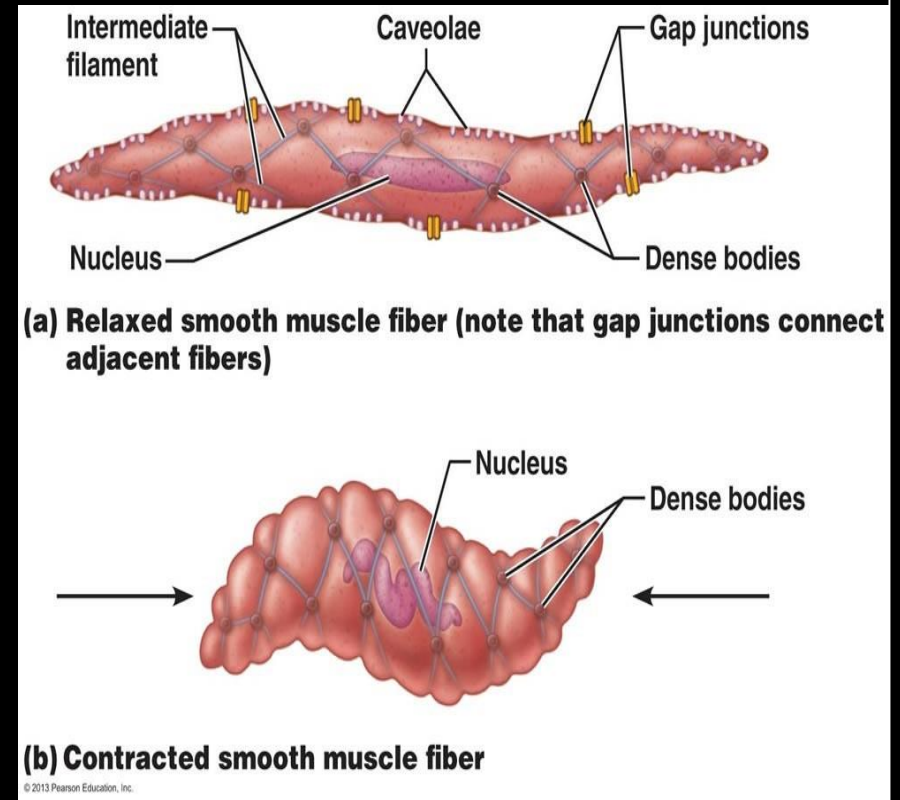
- These potentials **Summate** with repeated stimuli

- EJP & IJP** are local responses.



STRUCTURE OF SMOOTH MUSCLE FIBRE

- ✓ Spindle shaped cells with broad central part & tapering ends.
- ✓ Length & diameter varies with organ
 - ✓ **GIT** – 30-40 mm/5 mm diameter.
 - ✓ **Blood vessels** – 15-20 mm/2-3 mm.
 - ✓ **Uterus** – 300 mm /10 mm

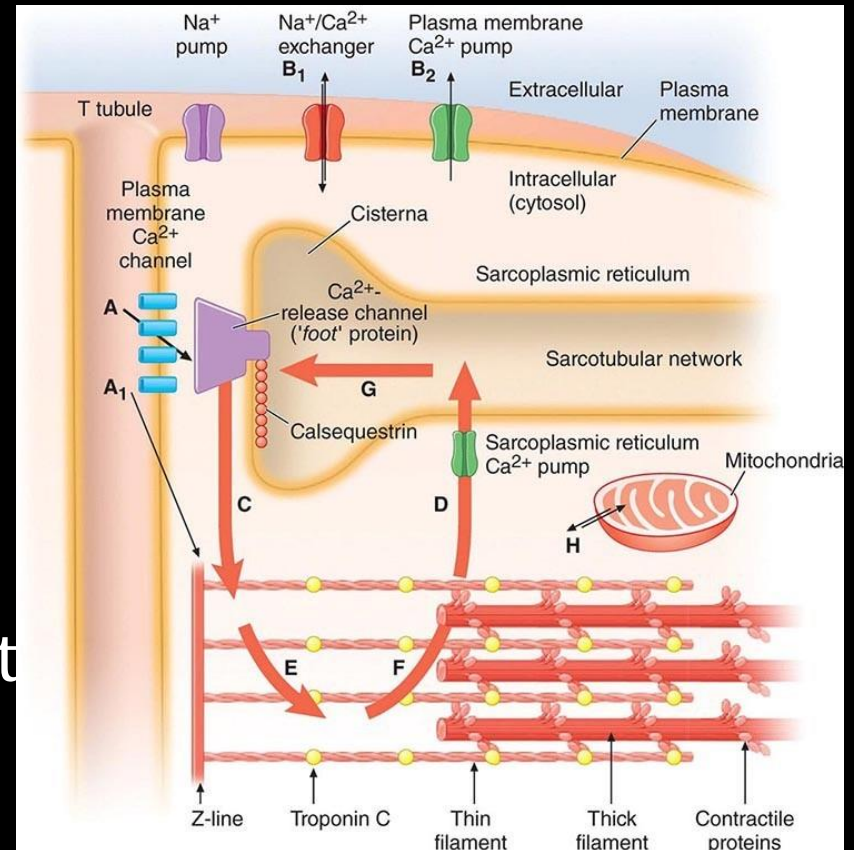


SALIENT FEATURES OF STRUCTURE OF SMOOTH MUSCLE FIBRE

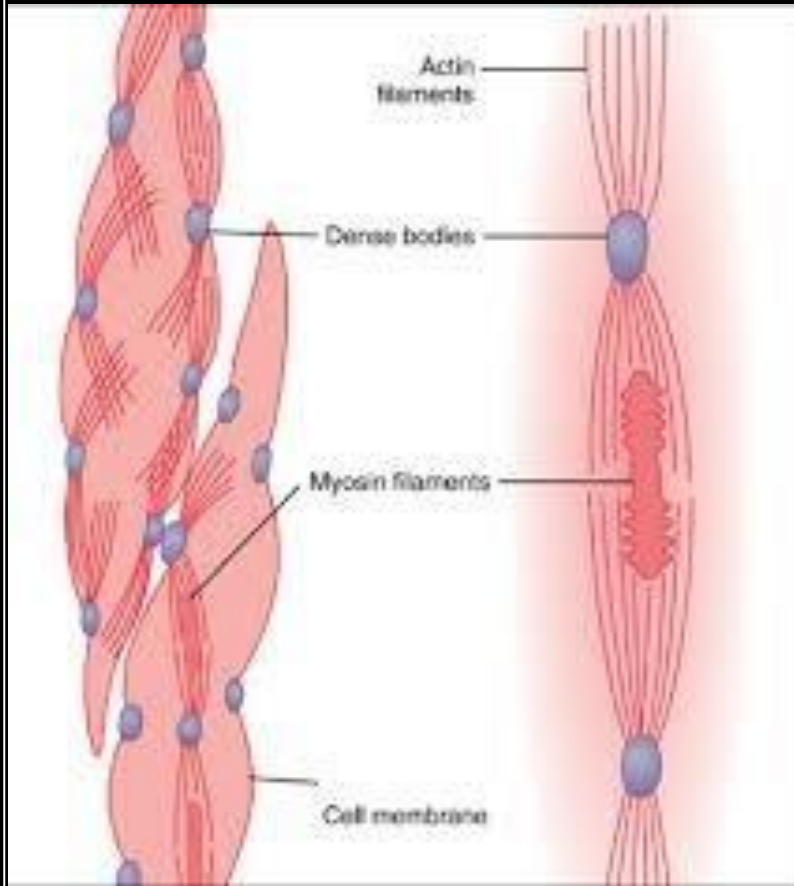
- ✓ **Plasma membrane** – surrounded by **External Lamina**.
 - ✓ Cell communicate through Gap Junction.
- ✓ **Sarcoplasm**
 - ✓ **Nucleus** – Central/Oval
 - ✓ Contains Mitochondria, Golgi apparatus, Endoplasmic Reticulum, Ribosomes.
- ✓ **Sarcoplasmic reticulum** – similar to skeletal muscle but not as developed.

SALIENT FEATURES OF STRUCTURE OF SMOOTH MUSCLE FIBRE

- ✓ **Myofibrils**- Sarcotubular system & triad not well developed.
- ✓ **Less** thick filaments & **More** thin filaments.
- ✓ Z line not well developed.
- ✓ **Actin** – Troponin is absent
- ✓ **Myosin** – bind only if phosphorylated.



SALIENT FEATURES OF STRUCTURE OF SMOOTH MUSCLE FIBRE



- ✓ **Dense bodies** – attached to the cell membrane
- ✓ Actin filaments are attached to dense bodies.
- ✓ When muscle contracts dense bodies are drawn close to each other.

PROCESS OF EXCITABILITY & CONTRACTILITY.

- ✓ **PROCESS OF MUSCLE EXCITATION**
- ✓ **PROCESS OF EXCITATION CONTRACTION COUPLING**
- ✓ **PROCESS OF MUSCLE CONTRACTION.**

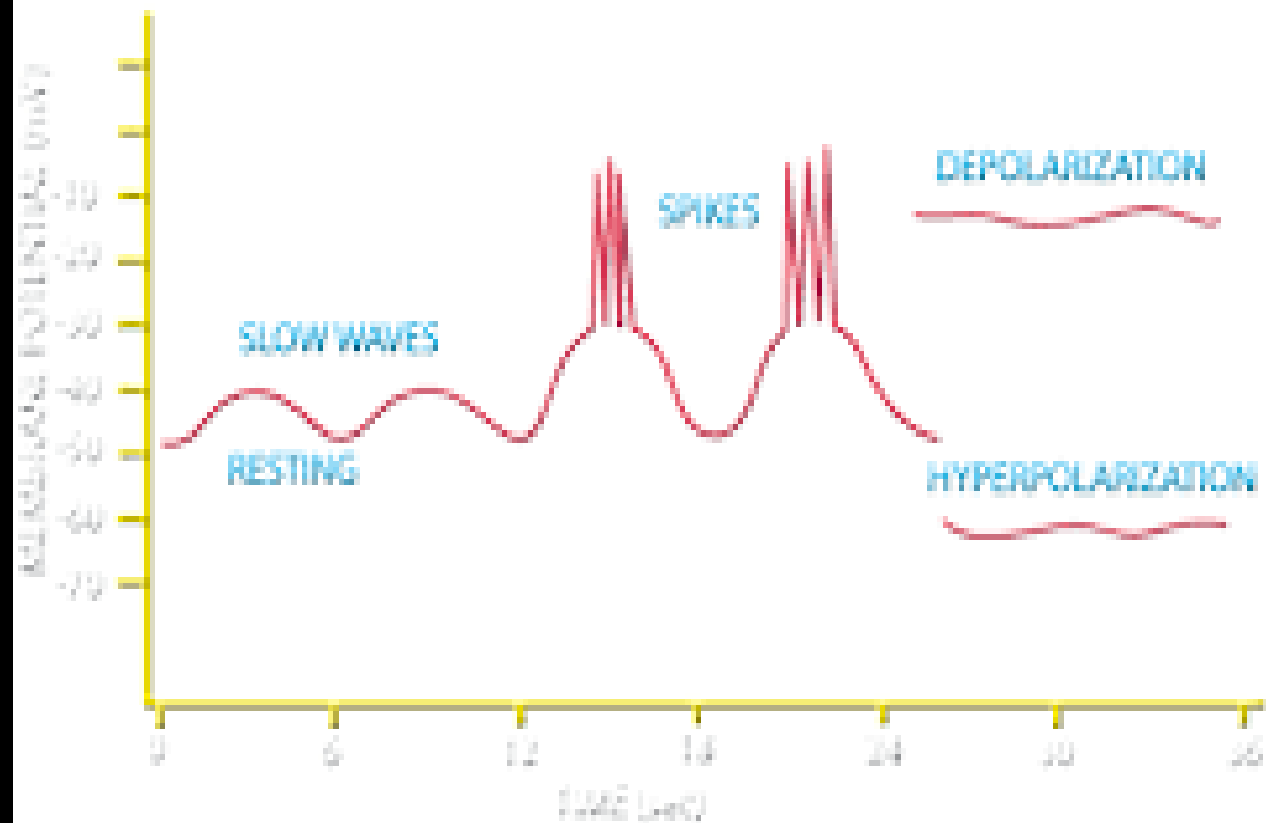
PROCESS OF MUSCLE EXCITATION

- ✓ **Electrical activity in single unit (Visceral) smooth muscles.**
 - ✓ Resting membrane potential
 - ✓ Action potential
 - ✓ Spike potential
 - ✓ Spike potential superimposed over slow wave potentials.
 - ✓ Action potential with plateau.

RESTING MEMBRANE POTENTIAL

- ✓ **Range between** -75mv to -90mv
- ✓ **Pe** kept to -35 mv.
- ✓ **Oscillations due to** Superimposition by pacemaker potential due to rhythmic change in **Ca channel permeability** & activity of Na-K Pump.

MEMBRANE POTENTIAL

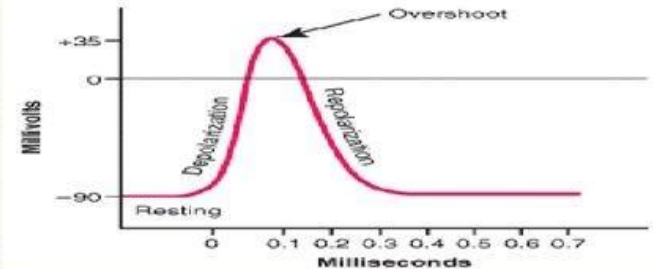


ACTION POTENTIAL

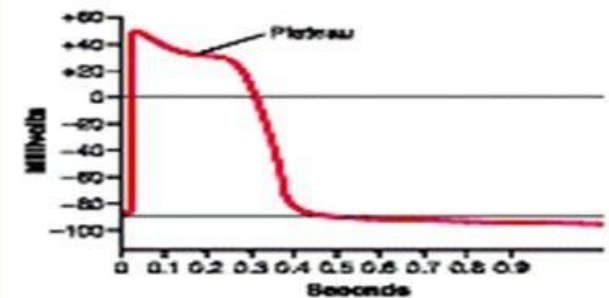
- ✓ When depolarization reaches threshold action potential begins & transmitted to other muscle cells through Gap junctions.
- ✓ **3 types action potential in general**
 - ✓ Spike potential
 - ✓ Spike potential over pacemaker potential .
- ✓ Action potential with plateau

Types & duration of AP

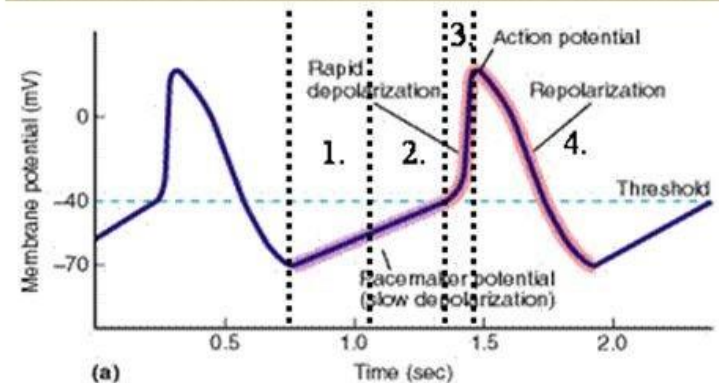
1. **Spike potential** (Nerve fibre, skeletal muscle) – 10 to 50m sec



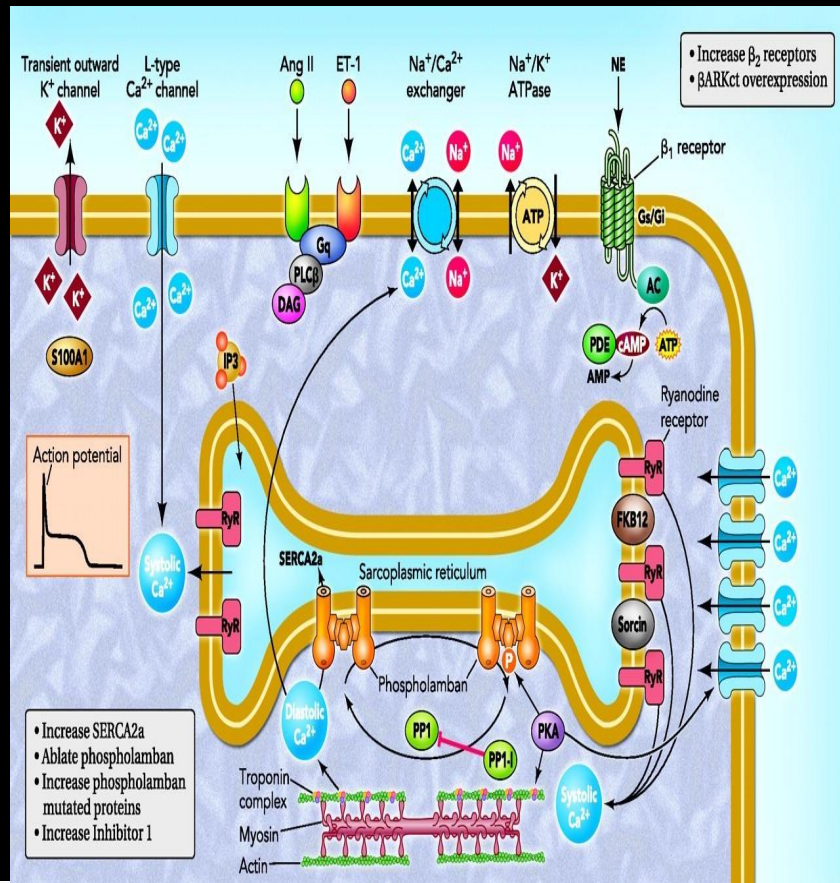
2. **Plateau type** (Myocardial cell & smooth muscle cell) – 250 to 350m sec



3. **Pace maker type**
(Conducting system of heart & smooth muscle fibres) – 100 to 150m sec



Ionic basis of action potential.

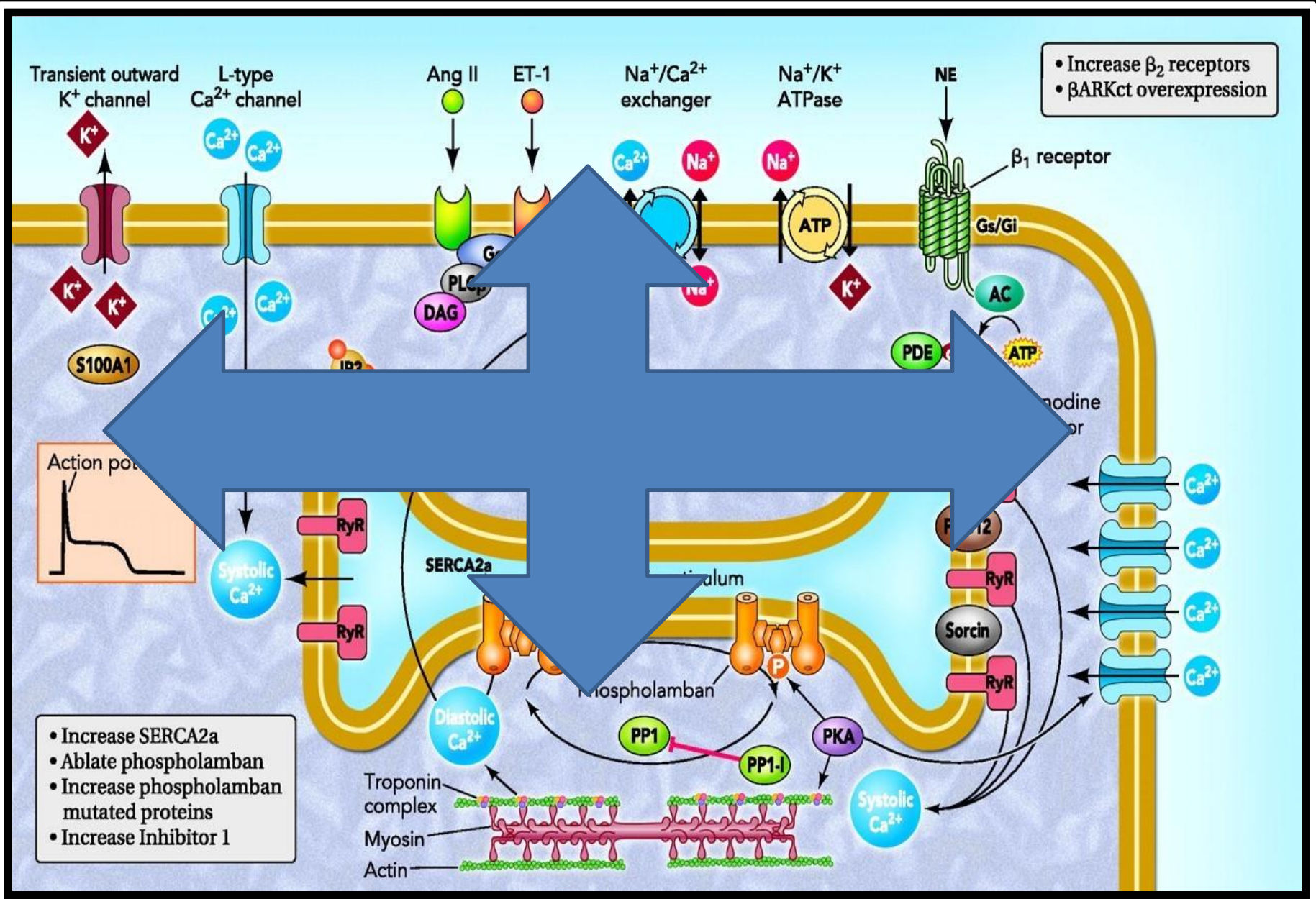


Depolarization – due to entry of **Ca** rather than Na.

This muscle has more voltage gated Ca channels than Na channels.

These open & close slowly & responsible for prolonged action.

Ca along with depolarization **also causes contraction** of smooth muscle.



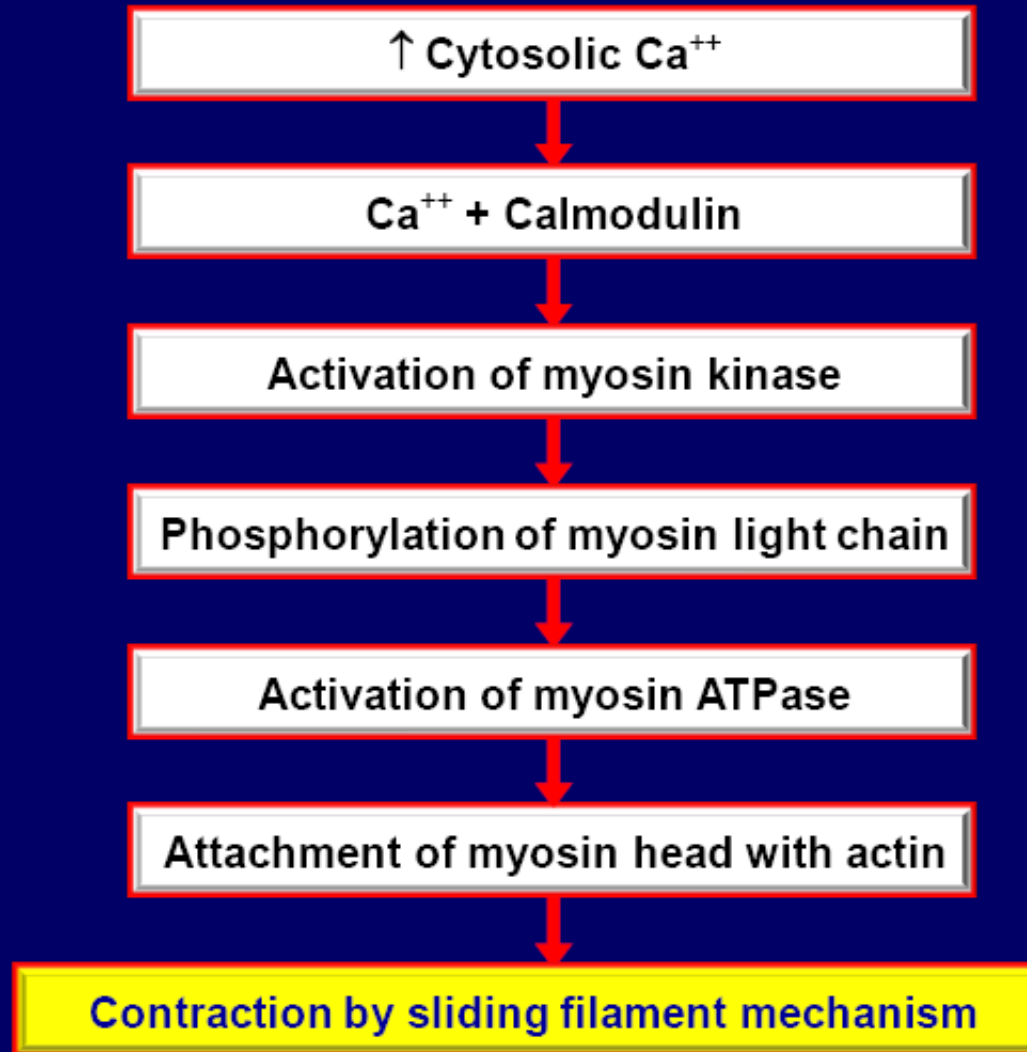
Electrical activity in multiunit smooth muscles.

- ✓ **Multi-unit smooth (Iris & Piloerector)** muscle respond to nerve stimulation which releases Ach & NE.
- ✓ These do not generate AP but causes excitatory Junctional Potential (EJP) & spread to entire fibre.

PROCESS OF MUSCLE CONTRACTION.

- ✓ Molecular mechanism is **similar** to skeletal muscle but..
- ✓ Smooth muscle don't contain Tropomyosin & Troponin.
- ✓ Light chain of Myosin acts as Tropomyosin & called **Regulatory** chain of Myosin.
- ✓ Ca binding protein **Calmodulin** acts as Troponin.

Mechanism of smooth muscle contraction



STEPS OF CROSS-BRIDGE CYCLING.

- ✓ **Ca combine with Calmodulin** – forms complex – **activates enzyme Myosin Light Chain Kinase (MLCK)**
- ✓ Phosphorylation of the Myosin Regulatory chain.
- ✓ Myosin head acquires capability to bind with actin & forms Cross-bridging.

STEPS OF CROSS-BRIDGE CYCLING.

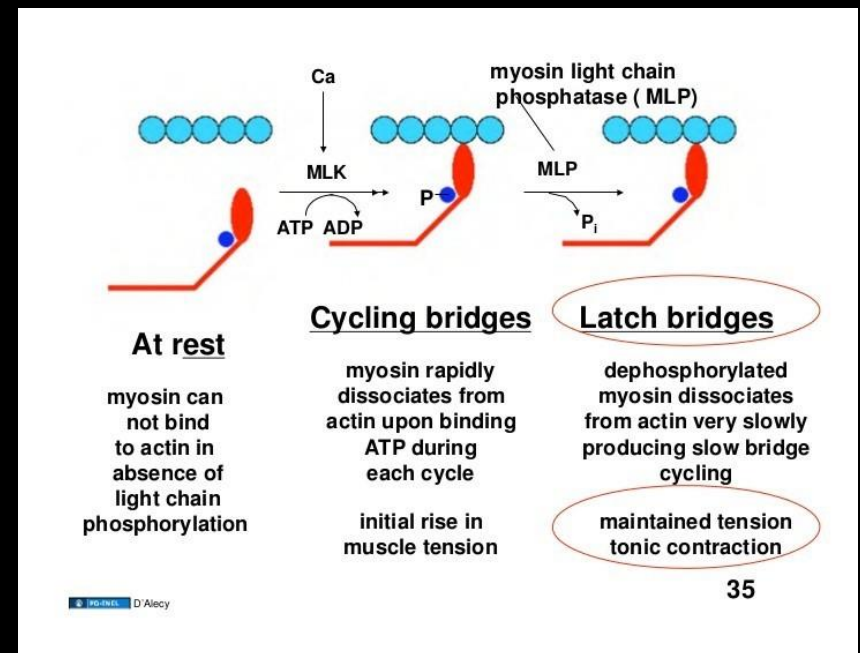
- ✓ **Power stroke** – formation of Actin-Myosin ADP Pi complex – conformational change in myosin head- flex towards arm – generate mechanical force i.e. Power Stroke.
 - ✓ Actin slide over myosin & causes contraction
 - ✓ Dense bodies are same as Z line.
- ✓ **Relaxation of smooth muscle** – Ca pump removes Ca from ICF to ECF – reverse all stages except Phosphorylation of Myosin.

CHARACTERISTICS OF EXCITABILITY & CONTRACTILITY.

- ✓ Slow excitation-contraction coupling.
- ✓ Plasticity.
- ✓ Latch phenomenon.
- ✓ Marked shortening of smooth muscle during contraction.
- ✓ Energy required to sustain smooth muscle contraction.

LATCH PHENOMENON.

- ✓ As this activation & cycling frequency decreases – **lower activation of enzymes** causes myosin head to remain attached to actin for longer time with less energy expenditure as ATP is required for detachment.



Energy required to sustain smooth muscle contraction.

- ✓ **Energy is much less** than skeletal muscle

EXCITATION & INHIBITION.

✓ **Excitation** –

- ✓ **Multi Unit Smooth Muscle** – stimulated through nerves
- ✓ **Single Unit Smooth Muscle** – through Nerves, Hormones, Pacemakers, Stretching, Cold Temperature.

FUNCTIONS

- ✓ Control movement of material through most of hollow organs
- ✓ Propel material in GIT
- ✓ Control blood flow in arterioles.
- ✓ Expell material from bladder.
- ✓ Control Piloerection
- ✓ Iris control.

INHIBITION OF SMOOTH MUSCLE

- ✓ **Through nerves by Sympathetic**
e.g. – Intestinal smooth muscles.
- ✓ **Through Hormones – Progesterone**
decrease activity of uterus.

*Thank
You*

